

Performance Test Result | ISM sponsors

1 message

Escotido, Paolo <escotidop@ismanila.org>
To: Renald Cruz <cruzs@ismanila.org>

ISM Sponsor

ISM Sponsor (#4)

10,000 VU . Apr 24, 2026 15:22:35 (5 mins) . Ramp up (4 minutes 20 seconds) . 32 logical cores . 61 GB total memory

VS

ISM Sponsor (#3)

5,000 VU . Apr 24, 2026 15:16:25 (5 mins) . Ramp u

Total requests sent ⓘ

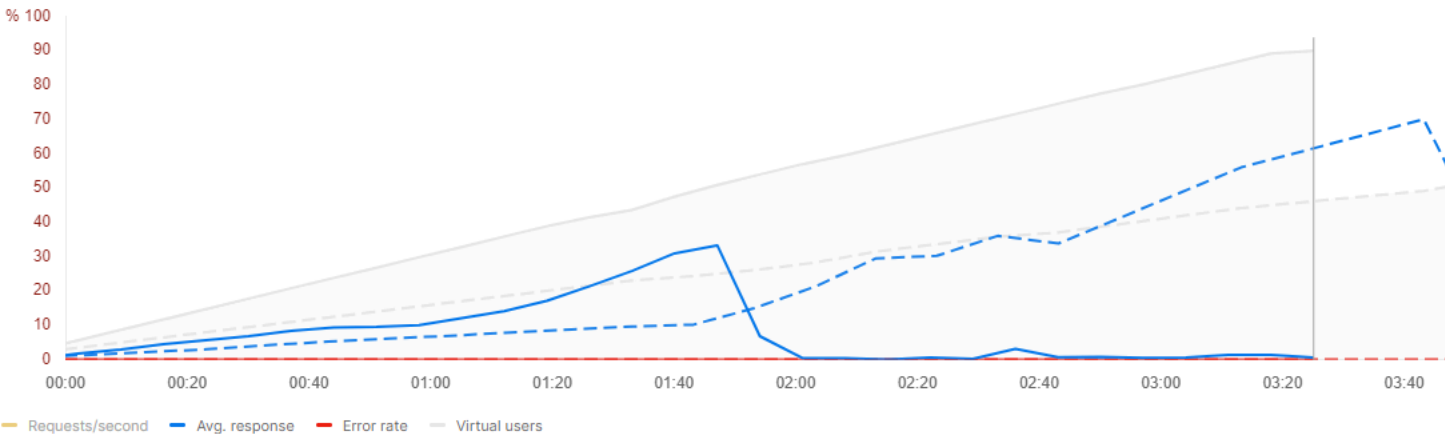
200,600 ▲ 129,200

Requests/second ⓘ

661.56 ▲ 423.97

Avg. response time ⓘ

2,159 ms ▼ 989



Performance details for total duration

Request	Total requests	Requests/s	Resp. time (Avg. r
GET Fetch sponsors	102,025 ▲ 65,693	336.47 ▲ 215.57	2,777 ▼ 6
GET Check Health	98,575 ▲ 63,507	325.09 ▲ 208.40	1,520 ▼ 1

Discussion

The performance test evaluated the capacity and stability of the ISM Sponsor API as an integration service intended for use by other systems that require access to ISM Sponsors data. The tests used a five-minute duration and a ramp-up period of four minutes and twenty seconds. This setup allowed the assessment to focus on how the API behaved as the number of simulated users increased.

The first test, which used 5,000 virtual users, recorded 71,400 total requests with a throughput of 237.59 requests per second. The average response time was 3,148 ms, and the error rate was 0%. The Fetch_sponsors endpoint handled 36,332 requests with an average response time of 3,467 ms, while the Check_Health endpoint handled 35,068 requests with an average response time of 2,777 ms. When the number of virtual users was doubled to 10,000, the total request count increased to 200,600, and the average response time decreased to 2,159 ms. The Fetch_sponsors endpoint processed 102,025 requests with an average response time of 2,777 ms, and the Check_Health endpoint processed 98,575 requests with an average response time of 1,520 ms.

The results suggest that the API demonstrated strong availability and reliability across both test scenarios. The absence of errors in both the 5,000 and 10,000 virtual user tests indicates that the API is capable of handling increased load without failures. This is important because the API is intended to support system integrations, where reliability is a key requirement. If external or internal systems depend on sponsor data, failed requests could lead to data inconsistencies or system downtime.

A notable finding is that the 10,000 virtual user test produced better overall performance than the 5,000 virtual user test. The total request count increased from 71,400 to 200,600, and the average response time decreased from 3,148 ms to 2,159 ms. This may suggest that the hosting environment, application service, or runtime resources performed better under increased load. It may also indicate that the service was able to make better use of available compute resources under increased load.

Despite these positive results, the maximum response times remain a concern. In the 5,000 virtual user test, the Fetch_sponsors endpoint reached a maximum response time of 170,670 ms. While most requests completed successfully, some requests experienced very long delays. For integration use, this is important because consuming systems may have timeouts that exceed the timeout threshold of the calling system.

The difference between average response time and maximum response time suggests that the API performance was generally stable but had occasional latency spikes. These spikes may be caused by database contention, or large response payloads. Since the Fetch_sponsors endpoint retrieves actual sponsor data, it is expected to be slower than the health check endpoint. However, the spikes in the maximum response time indicate that there are areas for optimization to ensure consistent performance under all conditions.

needed.

The health check endpoint also showed successful behavior in both tests. This confirms that the API service remained reachable during the test period. However, the health endpoint also recorded a 50% error rate. This may indicate that the service itself experienced temporary saturation, not only the sponsor data query. As a result, both application-level and infrastructure-level metrics should be reviewed.

Overall, the test results show that the ISM Sponsor API is capable of handling high concurrent traffic with no recorded request failures. From an availability perspective, the API passed both tests and showed strong improvement under the higher load condition. However, from a latency perspective, the API still requires optimization, especially for worst-case response times.

For an API integration that will be used by other systems, the results are acceptable but should not be treated as fully optimized. The 0.00% error rate is a strong indicator of reliability, but the results should be further reviewed for database indexing, query efficiency, caching, payload size, and Azure App Service scaling behavior. Monitoring should also focus on percentile-based metrics such as p99 and p999.

In conclusion, the ISM Sponsor API demonstrated reliable performance under both tested loads. It successfully handled 5,000 and 10,000 virtual users without errors. The higher-load test showed that the service can support large integration workloads. However, the presence of high maximum response times indicates that latency optimization is still necessary before the API is considered ready for production use.



Paolo Escotido
IT Development Engineer
International School Manila
www.ismanila.org +632.8840.8420

Notice: The contents of this email and any attachments are confidential. If you are not the intended recipient, please notify the sender and delete this message.

2 attachments

-  **ISM-Sponsor-performance-report-3.pdf**
2483K
-  **ISM-Sponsor-performance-report-4.pdf**
2503K